

Supplementary Materials

Robustness of the method: modeling activity with different levels of noise and detection of center of mass and trajectories

We evaluated the robustness of our method, i.e., that the operators responded to the distinctive physiological signals in the expected locations, by using a neuronal model. A standard model for a pyramidal neuron was chosen (Mainen & Sejnowski, 1996). The model (McDougal et al., 2017) was obtained from the Yale Medical Repository <https://modeldb.science/2488?tab=3>

The simulation for a single spike was run using the Python interface for NEURON with the default settings of the model (Hines, 1994), and the extra package LFPy, which calculates the external FP from the simulated activity of the neuron (Lindén et al., 2014). Then we added noise at various intensities in the following manner (Suppl. Fig. 2). We obtained the standard deviation from the recorded FP. We set four levels of noise, 0.05, 0.1, 0.5 and 1.0. We filled an array as large as the FP recording with Gaussian noise. Then we multiplied this noise signal by the standard deviation and by one of the levels of noise and added that to the original FP. We smoothed the signal with a Gaussian Blur filter and obtained the CSD by the difference method stated in the Methods section. The center of mass was obtained with disjoint components separated by four different parameters. Again, we used the standard deviation as our unit. The factors chosen were 0.25, 0.5, 1.5 and 3. We show these results in Supplementary Figure 2. The aim of this manipulation is to ascertain that the signals recorded from this “closed system” (i.e., the hippocampal slice), are stable and defined by the pure hippocampal system and are not

influenced by external signals, as may be cortical inputs. Thus, the interpretation is driven by what is directly observed, evidencing that the analysis method is robust and not disturbed by external (experimental noise in this case) signals (cortical influences in a slice experiment). We thus show that the trajectories show robust features against both noise and criteria for neutral set.

Gaussian Blur Denoising

Supplementary Figures 3 and 4 show two examples of the FP, the raw CSD representation and a CSD representation after a Gaussian Blur Filter has been applied, with a σ value of 42 μm (twice the inter-electrode distance). This acts as a form of spatial denoising mechanism, disclosing the disjoint components that are now apparent to the eye. It is interesting that sink/sources dipoles are apparent along the pyramidal cell layer. Those appearing in the proximal CA3 layer are long lasting. More sites of high current density appear toward CA1 in time (Suppl. Fig. 4). We further proceed to separate the disjoint components, as seen in Supplementary Figure 4 without and with the Gaussian Blur (GB). Even though well localized patches of activity are seen over the *stratum pyramidale* in the non-filtered signal, large parts of the activity remain hidden by the noise. After the Gaussian Blur has been applied the joint sets of activity are evident. This experiment served as a test of the applicability and robustness of our method. Although the signal was high and appeared to be appearing in multiple sites, the trajectories could be decomposed and the propagation of the burst of activity could be traced. The Gaussian blur filter not only deals with the background noise, but it even helps to delimitate the disjoint components. The noise averages to values well within the threshold for the separation of the components, and that pulls evanescent borders also inside this range.

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Supplementary Figure Legends

Supplementary Figure 1

Active Channel Detection. The procedure is thoroughly described in the “Data preprocessing and active channel detection” section. In short, we apply a thresholding algorithm to weight vectors built with the cardinality or the signal shift deviation to detect channels that have a high correlation with the anatomical distribution of the recorded tissue, or r.o.i., effectively discarding non-physiological channels. Thus, the aim is to identify electrodes (channels) most likely recording physiological signals, discarding channels exposed to saline solution or located outside the tissue. Panel A. Example of the detection method applied to a ventral rat hippocampal slice. Panel B. Same method applied to a rat striatal slice. Left column depicts the graphical representation of the weighted vector, derived either from the cardinality score, or from the signal shift deviation computed with a fixed time-delay window. The middle column displays thresholding scores, calculated by applying multiple thresholding algorithms to the weighted vector; the value at each channel reflects the number of times it exceeded a computed threshold (see Methods). The right column depicts the final binarized map of active channels, where minimal gaps are filled and isolated scattered detections are removed. This method successfully differentiates channels inside the tissue from those in the noise region, regardless of prior knowledge of the anatomy or the intrinsic activity level of the tissue.

Supplementary Figure 2

Centers of mass and trajectories obtained from a simulated field potential in a single-cell model. Modelled extracellular signal obtained from the simulated activity for a single spike from a single neuron with added Gaussian noise at 0.05X, 0.1X, 0.5X and 1X, times the standard deviation from the simulated extracellular signal (each row). The center of mass was obtained with disjoint components separated by 4 different thresholds: 0.25, 0.5, 1.5 and 3 times the standard deviation of the CSD.

The bottom two color maps validate the CM detection protocol using a synthetic CSD distribution generated via the kCSD-python package. This dipolar configuration (comprising a source and sink) was defined using `true_csd = CSD_PROFILE(csd_at, seed = 8)`. Applying the CM detection to this controlled distribution resulted in CMs that aligned closely with the geometrical centers of the true source and sink, confirming the method's fidelity.

Supplementary Figure 3

Gaussian blur filtering effects on the CSD

Color-coded representations of the FP and the corresponding CSD at different moments of the 4-AP-induced epileptiform activity. The first line of maps (FP) depicts raw, color-coded voltage measurements without filtering. The middle row (CSD) depicts the application of a difference operator. Such operators are overly sensitive to noise or abrupt change of values. Applying them directly to coarse valued data would result in

exaggerated or spurious edges for the sink and source sets. The third line of maps, CSD+GB, were obtained after applying a Gaussian spatial filter (Gaussian Blur). This filter considerably denoises the data, softens the boundaries of the sinks and source sets and uncovers the structure of the hippocampus due to the isolation of its activity from noise.

Supplementary Figure 4

The detection of disconnected components for the Sink and Source Sets.

Gaussian blur filtering of color-coded maps of the CSD. Although detected with a threshold value two times the standard deviation of the whole data in CSD representation (2α), the disconnected components for the sink and source sets cannot be clearly isolated because most of the joint activity gets hidden under the spatial noise (upper line of color maps, CSD+CM). The lower row depicts the detection of sinks and source sets after applying the Gaussian blur filter (GB+CSD+CM), where the components can be aggregated in sets, and can be distinguished from noise and interest sets. Larger coherent patches of activity are clearly shown, and the structure of the CA region is uncovered.